Lec 1 1.1-2.2

概念自检

- proposition
- **logical operator**
  - negation
  - conjunction
  - disjunction
  - exclusive or / XOR
- converse / inverse /contrapositive
  - $q \rightarrow p$
  - $\neg q \rightarrow \neg p$
  - $\neg p \rightarrow \neg q$
- truth table
- precedence of logical operators
- bitwise operations
- consistent
- **tautology / contradiction / contingency**
- logically equivalent
- logical equivalences
  - De Morgan's laws
  - Absorption laws
  - Implication law
  - Equivalence law
  - ……见附录
- satisfiable
  - 注意与 consistent 的区别
- predicate / propositional function
  - n-place predicate
- **quantifier**
  - universal quantification
  - existential quantification
  - bound variable
  - free variable
  - De Morgan's laws for quantifiers
  - Nested Quantifiers
- functionally complete

- **conjunctive normal form**
- **disjunctive normal form**
- **full CNF / DNF**
- **Prenex Normal Form**
- valid arguments
- rules of inference
    - modus ponens
    - modus tollens
    - resolution rule
    - ...for quantified statements
        - universal instantiation
        - universal generalization
        - existential instantiation
        - existential generalization
- *proof theorem axioms lemma conjecture
- methods of proofs
    - **proof by contraposition**
- Set
- Venn Diagram
- relations between two sets
- finite set
- **cardinality**
- **power set**
    - |S| = n -> |P(S)| = 2^n
- **Cartisian product**
- set operation
- set identity

习题整理

1. 2019春夏

1) ( ) "This statement is false." is a proposition.

2. llf 25 quiz1

( ) (1) The proposition "The volume of sun is smaller than that of earth only if there does not exist the four seasons on earth" is false.

3. zyw 25 quiz1

1.Are these system specifications consistent?

"The router can send packets to the edge system only if it supports the new address space.

For the router to support the new address space it is necessary that the latest software release be installed.

The router can send packets to the edge system if the latest software release is installed.

The router does not support the new address space."(10')

4. 2019秋冬

(e) ( ) If exactly one of the assignments 000, 011, 100, and 111 make the propositional formula φ false, then φ can be converted in full disjunctive normal form $\Sigma(0, 3, 4, 7)$.

5. llf 24 quiz1

$$(6)\ A \oplus (B \oplus C) = (A \oplus B) \oplus C$$

6. llf 25 quiz1

( ) (2) The proposition $((p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow r$ is a tautology.

7. llf 25 quiz1

(1) Let $F(A)$ be the predicate "A is a finite set" and $S(A,B)$ be the predicate "A is contained in B". Suppose the universe of discourse consists of all sets. Translate the statement into symbols.

Every subset of a finite set is finite.

8. turing 25 quiz1

2. (12%) Write a proposition equivalent to $p \oplus q$.

   a) Using only p, q, $\neg$, and the connective $\wedge$.

   b) Using only p, q, and the connective $|$ (where $|$ represents the NAND operation).

9. 2018秋冬

(e) ( ) Let $P(x)$, $Q(x)$ be two predicates, then $\forall x P(x) \to \exists y Q(y) \Leftrightarrow \forall x \exists y (P(x) \to Q(y))$.

10. 2019春夏

10) ( ) If $\forall x(P(x) \vee Q(x))$ and $\forall x((\neg P(x) \wedge Q(x)) \to R(x))$ are true, then $\forall x(\neg R(x) \to P(x))$ is also true, where the domains of all quantifiers are the same.

11. llf 25 quiz1 DNF

4.(10 points) Convert the following formula into logically equivalent formula in full disjunctive normal form. Determine whether it is tautology, contradiction or contingence. Find the assignments of p, q and r for which the formula is true.

$$\neg(r \to \neg q) \vee (p \wedge \neg r)$$

12. turing 25 quiz1 CNF

3. (9%) Find the full conjunctive normal form of $(p \oplus q) \vee r$.

13. zsy 25 quiz1

Determine whether each of the following formulae is true, if it is true, prove it; if not, give a counterexample. Where $P(x)$ and $Q(x)$ are predicates, the domain is positive integer.

a) $\forall x(P(x) \vee Q(x)) = \forall x P(x) \vee \forall x Q(x)$

b) $\exists x(P(x) \vee Q(x)) = \exists x P(x) \vee \exists x Q(x)$

14. llf 25 quiz1 universal/existential instantiation

5. (6 points) Use rules of inference to show that if $\forall x(p(x) \vee c(x))$, $\forall x(c(x) \leftrightarrow d(x))$ and $\exists x \neg d(x)$ are true, then $\exists x p(x)$ is true.

15. zsy 25 quiz1

Suppose that there are three types of people on an island, knights, knaves, and spies. Knights always tell the truth, knaves always lie, and spies sometimes lie and sometimes tell the truth. We know one of these people is a knight, one is a knave, and one is a spy. Each of the three people knows the type of person each of the other two is. For each of these situations, if possible, determine who the knave, knight, and spy are:

- A says "I am the knight," B says "A is not the knave," and C says "B is not the knave."
- What types of A, B, C?

集合定义与运算相关

16. turing quiz 1

If A, B, and C are sets, then

$$A - (B \cap C) = (A - B) \cup (A - C)$$

( )

17. 2018秋冬

(b) ( ) Let A, B be two sets. If $2^A \subseteq 2^B$, then $A \subseteq B$, where 2^X is the power set of X.

18. 2018秋冬

(a) ( ) There are two sets A and B, such that both $A \subseteq B$ and $A \in B$.

19. 2019春夏

( ) P(A) = P(B),if and only if A = B , where P(X) is the power set of X.

20. llf 25 quiz1

( ) (7) Suppose $A=\{a,\{a\}\}$, then we have $\{a\} \subseteq P(A)$

21. 2019秋冬

( ) Let A, B, C and D be arbitrary sets, then $(A \cup B) \times (C \cup D) = (A \times C) \cup (B \times D)$.

22. 书上习题改

设有集合A, B, C，如果$A \times B = A \times C$，是否有$B = C$

附录

证明方法相关（非重点，建议仅作了解）

题目： 请证明存在两个无理数 x 和 y，使得 x^y 是有理数。（非构造性存在性证明，非常巧妙）

Hint： 已知 $\sqrt{2}$ 是无理数。考虑数字 ……

题目： 在一个有 n（$n \geq 2$）个人的聚会上，证明至少有两个人认识的聚会人数一样多。（假设“认识”是相互的）

Hint： 使用**反证法**。假设所有 n 个人认识的人数都互不相同……

题目： 在一次单循环赛（每个人都与其他所有人比赛一场，没有平局）中，证明必定存在一名“国王”选手：他对于任何其他选手，要么直接击败了对方，要么击败过某个曾击败对方的选手。

Hint： 设赢下最多场比赛的选手为 A……

Rule of Inference（了解即可）

Rule of Inference	Tautology	Name
$\begin{array}{l} p \\ \underline{p \to q} \\ \therefore q \end{array}$	$(p \wedge (p \to q)) \to q$	Modus ponens 假言推理
$\begin{array}{l} \neg q \\ \underline{p \to q} \\ \therefore \neg p \end{array}$	$(\neg q \wedge (p \to q)) \to \neg p$	Modus tollens 拒取式
$\begin{array}{l} p \to q \\ \underline{q \to r} \\ \therefore p \to r \end{array}$	$((p \to q) \wedge (q \to r)) \to (p \to r)$	Hypothetical syllogism 假言三段论
$\begin{array}{l} p \vee q \\ \underline{\neg p} \\ \therefore q \end{array}$	$((p \vee q) \wedge \neg p) \to q$	Disjunctive syllogism 析取三段论
$\begin{array}{l} \underline{p} \\ \therefore p \vee q \end{array}$	$p \to (p \vee q)$	Addition 附加
$\begin{array}{l} \underline{p \wedge q} \\ \therefore p \end{array}$	$(p \wedge q) \to p$	Simplification
$\begin{array}{l} p \\ \underline{q} \\ \therefore p \wedge q \end{array}$	$((p) \wedge (q)) \to (p \wedge q)$	Conjunction

Rule of Inference	Tautology	Name
$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution 归结

Rule of Inference	Name
$\begin{array}{l} \forall x P(x) \\ \hline \therefore P(c) \end{array}$	Universal instantiation 全称指定
$\begin{array}{l} P(c) \text{ for an arbitrary } c \\ \hline \therefore \forall x P(x) \end{array}$	Universal generalization 全称推广
$\begin{array}{l} \exists x P(x) \\ \hline \therefore P(c) \text{ for some element } c \end{array}$	Existential instantiation 存在指定
$\begin{array}{l} P(c) \text{ for some element } c \\ \hline \therefore \exists x P(x) \end{array}$	Existential generalization 存在推广

扩展：功能完备的 (functionally complete)逻辑运算符集（非重点，建议仅作了解）

首先$\{\neg, \wedge, \vee\}, \{\neg, \wedge\}, \{\neg, \vee\}, \{|\}, \{\downarrow\}$ 均可为全功能集。

逻辑运算符集 $\{\neg, \rightarrow\}$ 是否是功能完备的。

答案： 已知 $\{\neg, \vee\}$ 是完备集。因为 $p \rightarrow q \equiv \neg p \vee q$，将等式中的 p 替换为 $\neg p$ 可得 $\neg p \rightarrow q \equiv p \vee q$。即仅用 $\{\neg, \rightarrow\}$ 即可构造出 $\vee$，因此该集合功能完备。

判断逻辑运算符集 $\{\wedge, \vee\}$ 是否功能完备。

答案： 不完备。因为 $\wedge$ 和 $\vee$ 具有保真性（即输入全为真 T 时，输出必为真 T），无论如何嵌套都无法在全真输入时输出假 F，因此无法构造出逻辑非 $\neg$。

引入逻辑假常量 $\perp$，证明 $\{\rightarrow, \perp\}$ 是功能完备的。

答案： 已知 $\{\neg, \rightarrow\}$ 是完备集。利用常量 $\perp$ 构造 $\neg$：$p \rightarrow \perp \equiv \neg p \vee \perp \equiv \neg p$。现已用 $\{\rightarrow, \perp\}$ 构造出 $\neg$，且原集合包含 $\rightarrow$，故该集合完备。

逻辑运算“非蕴含”定义为 $p \nrightarrow q \equiv p \wedge \neg q$。引入逻辑真常量 $\top$，证明 $\{\nrightarrow, \top\}$ 是功能完备的。

答案： 已知 $\{\neg, \wedge\}$ 是完备集。构造 $\neg$：$\top \nrightarrow p \equiv \top \wedge \neg p \equiv \neg p$。构造 $\wedge$：$p \nrightarrow (\neg q) \equiv p \wedge \neg(\neg q) \equiv p \wedge q$。两者均可被构造，故完备。

判断集合 $\{\rightarrow, \nrightarrow\}$ 是否是功能完备的。

答案： 已知 $\{\neg, \rightarrow\}$ 为完备集。首先构造常量 $\top$：$p \rightarrow p \equiv \top$。接着利用 $\top$ 和非蕴含构造 $\neg$：$\top \nrightarrow p \equiv \top \wedge \neg p \equiv \neg p$。因为已构造出 $\neg$，且集合包含 $\rightarrow$，故完备。

判断逻辑运算符集 $\{\neg, \oplus\}$ 是否为功能完备集。

答案： 不完备。在模2算术中，异或 $\oplus$ 对应加法，非 $\neg p$ 对应 $p \oplus 1$。两者均为线性运算（仿射变换）。仅利用线性运算无法构造出非线性运算（如逻辑与 $\wedge$ 对应的模2乘法 $p \cdot q$），故无法表达所有逻辑函数。

判断逻辑运算符集 $\{\rightarrow, \leftrightarrow\}$ 是否为功能完备集。

答案： 不完备。$\rightarrow$ 和 $\leftrightarrow$ 均具有保真性。当所有输入变元为真时，$T \rightarrow T \equiv T$ 且 $T \leftrightarrow T \equiv T$。因此无论如何嵌套，都无法在全真输入时得到假的结果，从而无法构造出逻辑非 $\neg$。

- 通用方法：波斯特完备性定理 (Post's Completeness Theorem)（感兴趣可自行搜索）